

Overview

Conclusions of a five-class book bot that predicts a sentence's author based on its style are presented in this article. TensorFlow/Keras is used for sequence modeling, while Spark is used for data loading, cleaning, and exploratory data analysis in the final process (TensorFlow, n.d.). There were 45 columns and 115,471 accurately processed entries in the dataset utilized for this task. 115,430 clean records remained after 41 duplicate sentence-author pairs were eliminated. A balanced modeling sample of 50,000 phrases was constructed following an exploratory investigation of the entire set of cleaned data. Both a baseline LSTM and an improved Bidirectional LSTM were trained. The improved model achieved a best validation accuracy of 70.00%, compared with 67.58% for the baseline model, and achieved a final test accuracy of 68.66% on the untouched 10,000-sentence test set.

A functional closed-set classifier for Mark Twain, Herman Melville, Jane Austen, Louisa May Alcott, and Charles Dickens is the end result. The findings demonstrate that it is possible to learn and anticipate a sentence's author based on sentence-level style characteristics.

1. Dataset Suitability

The selected open dataset includes sentence-level writings from five different nineteenth-century writers (Rezaei, 2025). It contains both precomputed language characteristics and raw sentences. The phrase itself is the primary predictive feature, which makes it suitable for LSTM. The author may be deduced from the word order and context, and an LSTM layer is good at sequence prediction and retains knowledge about prior words. Additionally, since each phrase has five potential labels (authors) to predict, the job is multi-class classification (Graves, 2012).

LINK TO DATASET - <https://huggingface.co/datasets/Mosab-Rezaei/19th-century-novelists/tree/main>

The revised spark load produced 115,471 rows with 45 columns, far more than the assessment's minimum requirement. Before duplicates were eliminated, the five authors were represented by 19,033 to 26,856 sentences, which is sufficient for both model training and exploratory analysis. As a result, the dataset is both manageable in a local development context and large enough to support neural sequence modeling.

The size of the dataset did not determine its eligibility for analysis. First, missing values, empty text fields, duplicate sentence-author data, class imbalance, excessive sentence length, label leakage, and linguistic differences were all examined. The five uniquely encoded author fields, Label_(number), Label_(tuple), and Merged are examples of columns with precise label-encoding fields that were recognized as leakage concerns and removed from the LSTM input. This was significant because, in the absence of writing style training, any of these variables may provide the model with the correct answer. However, the dataset does not perfectly capture the style of authors. A neural network can acquire recurring names, places, subjects, or book-specific terminology in addition to just style data, since sentences are derived from books. Results should not be generalized to random current writers because the data also covers a closed five-author set and nineteenth-century English. Rather than being seen as model failures, these restrictions are taken into consideration when analyzing the final model.

2. Exploratory Data Analysis

2.1 Missing Values and Duplicates

There were 115,471 entries in the source. Neither Sentence nor Author had any missing values, and there were no sentences that included blanks. Nevertheless, 41 Sentence-Author pairings were identical. There were 115,430 clean observations for modeling once these duplicates were eliminated. The likelihood of the same identical text appearing in both training and testing sets is reduced by eliminating duplicates prior to any split.

Quality check	Result	Action
Missing Sentence values	0	Nothing Required
Missing Author values	0	No rows removed
Blank sentences	0	No rows removed
Duplicate Sentence-Author pairs	41	Removed before EDA
Clean records	115,430	Used for full EDA

2.2 Class Distribution

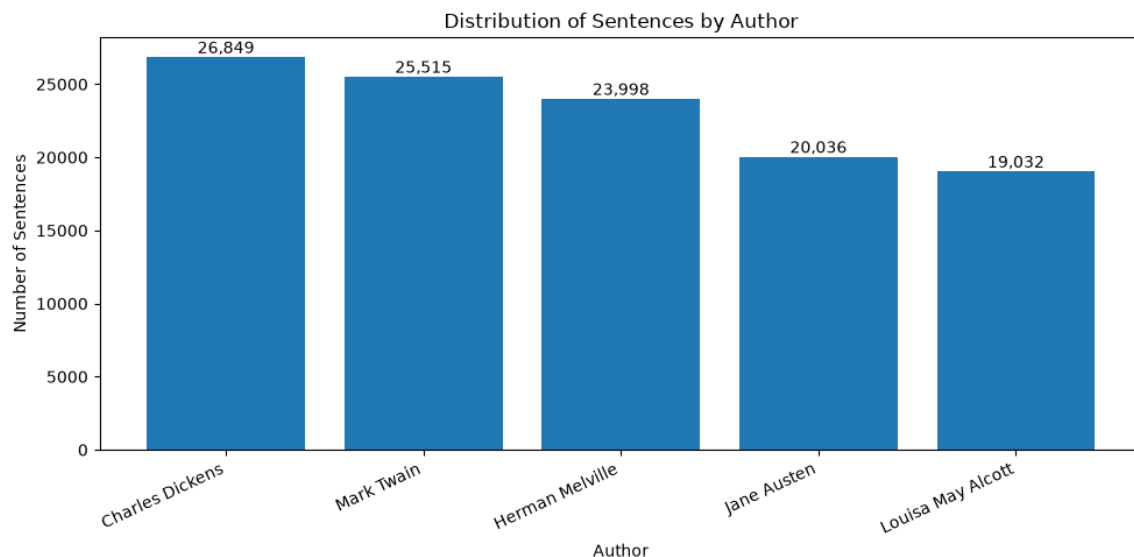


Figure 1 - Distribution of Sentences – Grouped by Author

At 26,849 sentences (23.26%), Charles Dickens accounted for the biggest portion of the cleaned dataset, followed by Mark Twain at 25,515 (22.10%), Herman Melville at 23,998 (20.79%), Jane Austen at 20,036 (17.36%), and Louisa May Alcott at 19,032 (16.49%). The more occurrences for the most represented author might still have an impact on a neural classifier, even if the imbalance is moderate rather than severe.

To maintain equilibrium, 10,000 phrases were chosen for each author using a Spark window with random seed 42. To avoid class frequencies becoming a simple shortcut for the model, the final modeling dataset included precisely 50,000 observations.

2.3 Sentence Length

Sentence length was analyzed since it directly affects tokenization, padding, and computing performance. In particular, the mean sentence length for the 115,430 observations following data cleaning was 22.68 words, with a standard deviation of 12.6. The sentence length varied from 1 to 83 words, with a positive skewness of 0.597, indicating a minor asymmetry of the distribution.

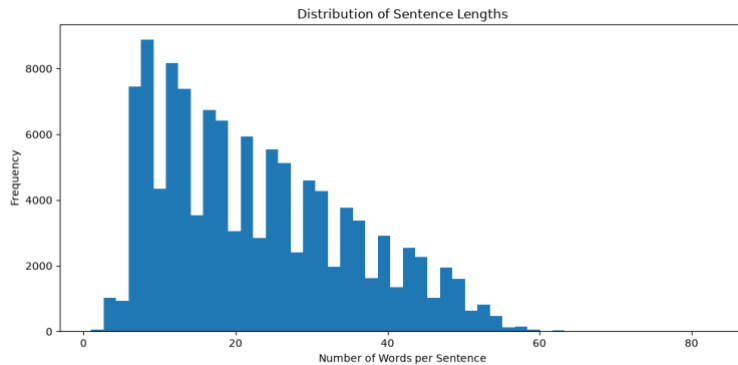


Figure 2 - Distribution of Sentence Length

There were some variations among the writers, according to the author-level analysis. Alcott had the longest average sentence length (24.91 words), followed by Herman Melville (24.42). Dickens received 21.41, Austen 21.03, and Twain 22.00. As a result, the observed patterns supported the original theory that the writers may exhibit a certain number of distinctive qualities.

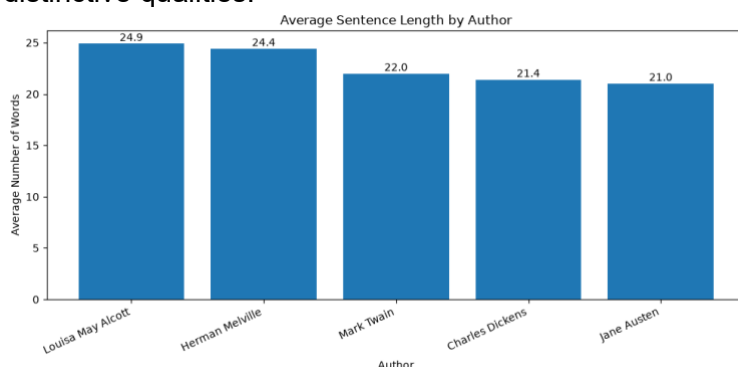


Figure 3 - Average sentence length by Author

2.4 Outliers

The interquartile range was used to identify outliers. With 12 words for Q1 and 31 words for Q3, the calculated value is 59.5 and the IQR is 19. None of the sentences fell below the bottom threshold, which is -16.5, however 98 sentences exceeded this range. All outliers were left in the dataset as it was discovered that the longest samples contained legitimate literary sentences. Although the proportion of outliers for each writer is less than 0.21%, Herman Melville's outliers reached 50, or 0.21% of all his data, which is greater than 0.07% for other authors. In order to maintain stylistic diversity and avoid excluding any authors on the same curve, it is helpful to leave outliers in the data.

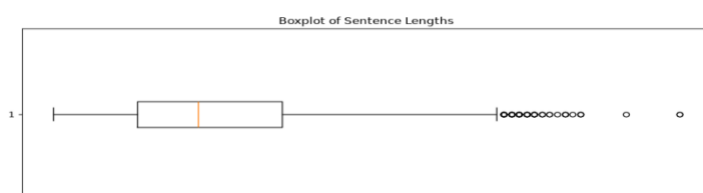


Figure 4 - Sentence Lengths Boxplot

Author	Total	Long_Sentences	Long_Sentence_Percentage
Herman Melville	23998	50	0.21
Mark Twain	25515	19	0.07
Charles Dickens	26849	20	0.07
Louisa May Alcott	19032	5	0.03
Jane Austen	20036	4	0.02

Figure 5 - Percentage of long sentences by Author

2.5 Linguistic patterns

Both raw sentence text and constructed linguistic characteristics are included in the dataset. Instead of using features as direct LSTM inputs to the model, EDA used features to identify author-level differences. Verb, adverb, noun, coordination and subordination percentages, pronoun percentages, prepositional phrase measures, noun phrases and subject counts were used to compare author groups.

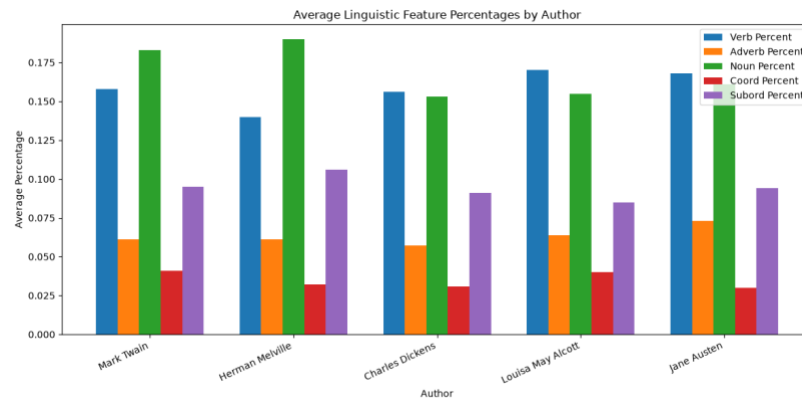


Figure 6 - Linguistic feature percentages

There were some differences in style. The greatest average adverb proportion (0.073) was found in Jane Austen. The greatest noun percentage (0.190) and subordination percentage (0.106) belonged to Herman Melville. Louisa May Alcott had average verb percentage (0.170). At 10.08, Charles Dickens had the lowest average parse-tree depth. Though these differences by themselves cannot confirm authorship, taken as a whole, they may suggest that the sequential text is a signal that can be learned.

3. Feature Selection

Since there were many columns in the original dataset that were helpful for analysis but not suitable as model inputs, feature selection was especially crucial in this project. The final input was the sentence content, and the target variable was the author's name because the model was designed to predict the author from the writing.

A number of columns that would have introduced target leakage and directly revealed the solution were purposefully eliminated. These included `Label_(number)`, `Label_(tuple)`, `Merged`, and the separate one-hot author columns for Charles Dickens, Jane Austen, Mark Twain, Louisa May Alcott, and Herman Melville. The model could have learned the authors from the labels in the data itself if these variables had been included.

Numerous computational language characteristics, including the percentage of verbs, nouns, adverbs, coordination, subordination, and pronoun usage, were also included in the dataset. Although they weren't used as LSTM inputs, these were helpful during EDA since they showed quantifiable variations in authors' writing styles. It seemed most logical to utilize the raw cleaned sentence as the feature of interest as the LSTM was designed to learn sequential patterns from the sentence text.

Sentence (predictor) and Author (target) were thus the only two variables of relevance in the modeling dataset. The phrases were transformed into padded number sequences and given to the neural network once the text had been cleaned and tokenized. This feature selection strategy prevented data leakage and redundancy while keeping the model concentrated on the job of identifying authorship based on stylistic elements of the text.

4. Text Preprocessing and Modelling Dataset

50 000 records were produced using a sample of 10 000 phrases per author. The Keras Tokenizer was used to preprocess the text where punctuation and special characters were removed, and the text was changed to lowercase. The model was then trained using the tokenized text.

Preparation choice	Final setting	Rationale
Initial split	80/20 split	Creates an 10 000 test set
Validation	20% of training	64% train, 16% validation, 20% test.
Vocabulary	5,000 most frequent words	Keeps a broad but bounded vocabulary.
Maximum sequence length	50 tokens	All tokenised training sentences were 50 tokens or fewer.
Padding / truncation	Post-padding / post-truncation	Creates fixed-size inputs for the recurrent network.

There were 40,000 training and 10,000 testing records, with 8,000 training and 2,000 testing examples per class. About 32000 records were used and 8000 records were used to assess the performance of the model since.

To prevent vocabulary information from the test set from being exposed to the training process, the tokenizer was only fitted on X_train. Sequence-length statistics analysis of the training text had a mean of 20.74 tokens, median of 19, maximum of 45, and 99th percentile of 44.

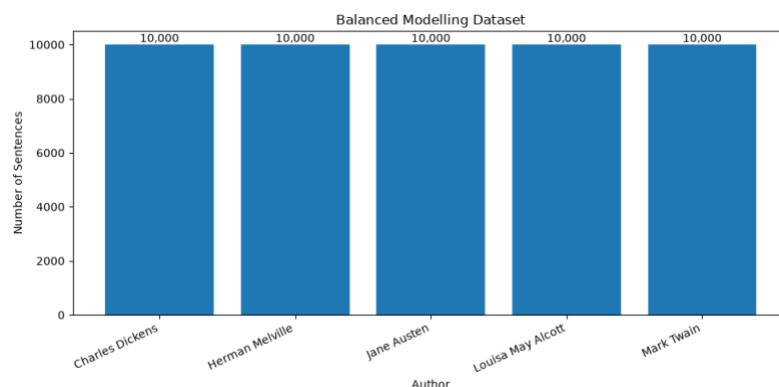


Figure 7 - Balanced Modelling Dataset

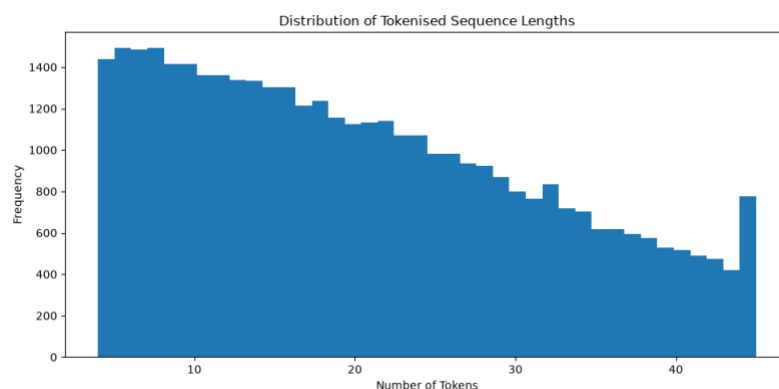


Figure 8 - Tokenized Sequence Lengths

5. Baseline LSTM

5000x64 embedding layer, one 64-cell LSTM layer, and a five-cell softmax output layer (a total of 353,349 trainable parameters) made up the initial network design, which was purposefully kept simple to enable subsequent modifications to be evaluated in relation to a basic comparative baseline. Thus, The five mutually conflicting integer-encoded output classes required the use of sparse categorical cross-entropy loss.

Component	Configuration
Embedding	Input vocabulary 5,000
Recurrent layer	LSTM with 64 units
Output	Dense(5), softmax
Loss	Sparse categorical cross-entropy
Optimizer	Adam
Batch size	64
Maximum epochs	10
Early stopping	Monitor val_loss, patience=2, restore best weights

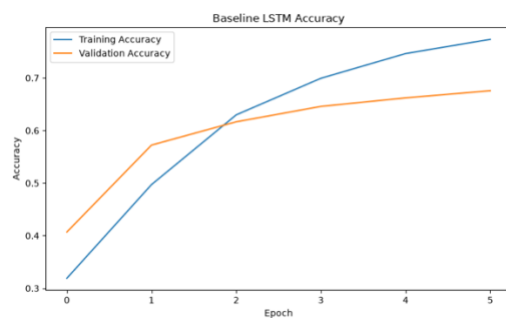


Figure 9 - Baseline Accuracy

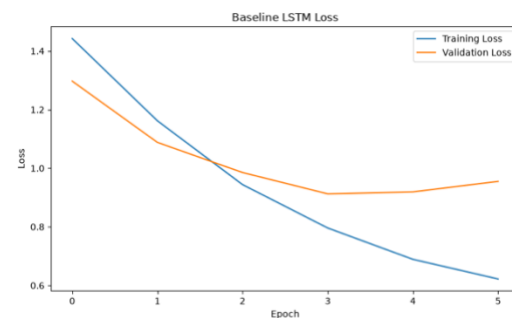


Figure 10 - Baseline Loss

The baseline had the best validation loss of 0.9125 and the greatest validation accuracy of 0.6758. While the validation accuracy remained in the mid-0.60s, the training accuracy increased further, reaching about 0.78. Training loss continued to decline as validation loss improved until the middle of training, at which point it halted. The network was improving at categorizing the training instances, but it was not making the same progress on unseen validation material, which is a sign of overfitting.

As a result, the baseline served as a diagnostic model and proved that the text included learnable signal about the author. However, it also showed that the single-direction LSTM without regularization was not generalizing nearly as effectively as was required. These results led to a specific modification of the next model.

6. Improved bidirectional LSTM

A bidirectional recurrent layer with dropout regularization was employed in the second model. A respectable comparison at the text-representation level was possible since the embedding configuration was still set to 5,000 words and 64 dimensions. The sequence was read in both ways using a bidirectional 64-unit LSTM, which produced a 128-dimensional representation. A dropout of 0.4, a 32-unit ReLU dense layer, a dropout of 0.3, and the five-class softmax output came next. There were 390,341 trainable parameters in the enhanced network.

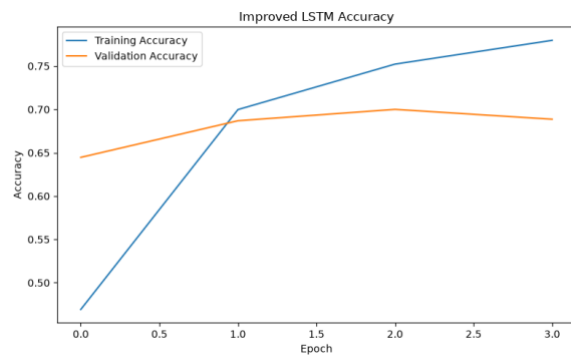


Figure 11 - Improved Model Accuracy

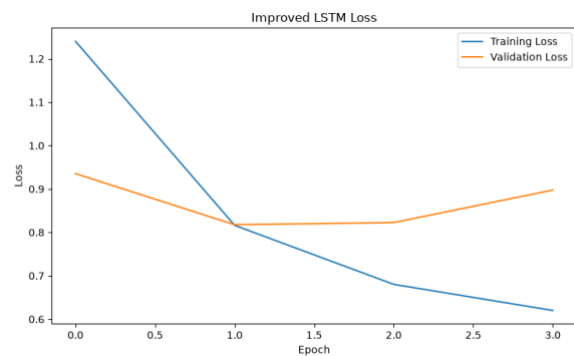


Figure 12 - Improved Model Loss

Model	Best validation accuracy	Best validation loss
Baseline LSTM	0.6758	0.9125
Improved Bidirectional LSTM	0.7000	0.8177

The improved model increased best validation accuracy from 67.58% to 70.00%, an improvement of approximately 2.43 percentage points. Best validation loss decreased from 0.9125 to 0.8177, a reduction of approximately 0.0948. As a result, both the optimization objective function and the classification metric improved, indicating adequate training progress. After achieving the optimal outcome at epoch 3, the validation loss started to rise while the training accuracy continued to rise. Early stopping was therefore advantageous and assisted in restoring the optimal weights associated with the lowest validation loss. Although overfitting was not completely eradicated, it was reduced, and the validation results were improved. For the last test on the unaltered test set, the best model was selected. It is the best practice to assess the final performance, and the test set was not utilized in the model selection process.

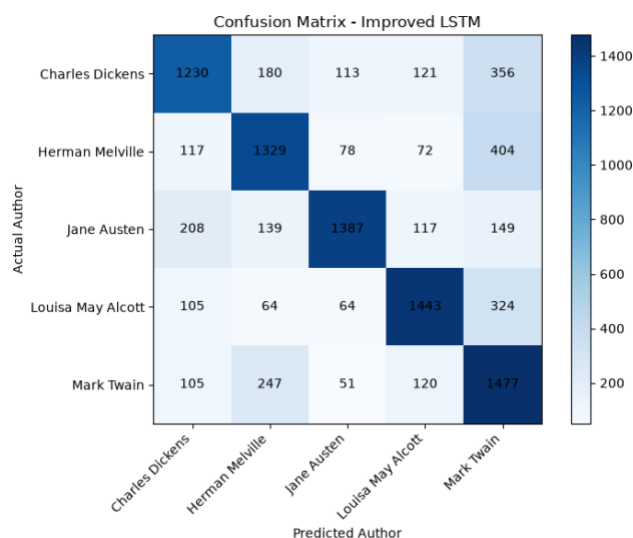
7. Final Evaluation

It was decided to use the upgraded Bidirectional LSTM for the last testing cycle after comparing the baseline and upgraded models on validation performance. The enhanced model raised the best validation accuracy from 67.58% (the baseline model) to 70.00%, according to an analysis of the accuracy data. Additionally, it was discovered that the enhanced model has lowered the best validation loss from 0.9125 (the baseline model) to 0.8177 when taking into account the loss findings. Therefore, it may be claimed that the updated design was selected for the final testing cycle because it performed better in generalizing to the unseen validation data (Scikit-learn, n.d.).

The untouched 20% test set, was then used to test the final model. Testing revealed that the model had a test loss of 0.8132 and a test accuracy of 68.66%. Therefore, one could claim that the model accurately identified about seven out of ten previously unknown authors of sentences. Beyond overall accuracy, the model's performance was examined using the classification report. The model's Weighted F1 score and Macro F1 score were both 0.6897. It is anticipated that the Macro and Weighted F1 scores, which show the average recall and precision of the model, will be the same because each author had an equal number of occurrences in the test set.

Author	Precision	Recall	F1-score	Support
Charles Dickens	0.697	0.615	0.653	2,000
Herman Melville	0.678	0.665	0.671	2,000
Jane Austen	0.819	0.694	0.751	2,000
Louisa May Alcott	0.770	0.722	0.745	2,000
Mark Twain	0.545	0.739	0.627	2,000
Overall / Macro	0.702	0.687	0.690	10,000

With an F1-score of 0,751, Jane Austen outperformed the others, while Louisa May Alcott came in second with 0,745. This indicates that the model performed better than the others in differentiating between these two authors. Mark Twain and Charles Dickens seemed to be the least effective at classifying. The model's shortcomings can also be shown using the confusion matrix.



These findings show that the model was successful in identifying the relevant patterns for each author's work, although some writers had enough similarities and language to make it challenging to distinguish between them at the sentence level. Nevertheless, for a five-class classification issue on individual phrases rather than passages, the ultimate accuracy rate of 68.66% is a rather good result.

Figure 13 - Confusion Matrix of improved model on test data

8. Book Bot

Creating a Book Bot that could determine which of the five writers is most likely to have authored a certain sentence was the project's last phase.

The training stage's preprocessing is carried out via the `predict_author()` function. The modified LSTM outputs a probability for each author and returns the one with the greatest probability after converting the input phrase to lower case, eliminating punctuation and special characters, and padding the resultant tokens to 50.

To demonstrate the bot, an actual sentence from the untouched test dataset was used:

“had rotted and died the french monarchy had gone down in the red tempest of the revolution and napoleon was a name that was beginning to be talked about”

The true author was **Mark Twain**, and the Book Bot also predicted **Mark Twain** with a confidence of **68.70%**.

```
BOOK BOT RESULT
-----
Predicted Author: Mark Twain
Confidence: 68.70%

Author Probabilities:
Charles Dickens: 10.11%
Herman Melville: 3.95%
Jane Austen: 1.05%
Louisa May Alcott: 16.19%
Mark Twain: 68.70%
```

Figure 14 - Book Bot Prediction

9. Conclusion

By using the relevant methods, it was possible to produce a working Book Bot comprising five authors. Spark analysis showed that the source data is sufficiently vast, comprehensive, and varied, and that the experiment's integrity was improved by eliminating duplicates and implementing target-leakage protections. Sentence length and linguistic patterns varied between authors during EDA. These results demonstrate the authors' unique, recognizable patterns and their contribution to the LSTM's sequence-length judgments.

A balanced 80/20 stratified train/test split, a 5,000-word training-only vocabulary and 50-token padded sequences were used in the final pipeline. The baseline LSTM achieved a best validation accuracy of 67.58%, while the improved Bidirectional LSTM achieved 70.00%. Best validation loss also improved from 0.9125 to 0.8177. The final model achieved a test accuracy of 68.66%, with a Macro F1-score of 0.6897.

Because sentence information, authorship is difficult to establish. Even after eliminating exact duplicates, phrases from the same novels may still exist in both the training and testing sets, potentially leading to target leakage. Rare words are assigned an out-of-vocabulary token by the 5,000-word restriction, and the model must always predict one of five authors, even for texts that are not authored by them. Additionally, the model should not be utilized as a general-purpose authorship classifier because the dataset only includes English literature from the nineteenth century.

Future research could examine Melville/Twain confusions, compare different vocabulary sizes and recurring unit sizes, experiment with paragraph- or multi-sentence level inputs, and, if book IDs become accessible, separate by source novel. Instead of modifying the present experiment, which aims to show a complete LSTM pipeline using the taught technique.

Bibliography

Graves, A., 2012. Long short-term memory. *Supervised sequence labelling with recurrent neural networks*, pp. 37-45.

Rezaei, M., 2025. *19th-century-novelists*. [Online]

Available at: <https://huggingface.co/datasets/Mosab-Rezaei/19th-century-novelists>

[Accessed September 2026].

Scikit-learn, n.d. *sklearn.metrics*. [Online]

Available at: <https://scikit-learn.org/stable/api/sklearn.metrics.html>

[Accessed September 2026].

TensorFlow, n.d. *LSTM*. [Online]

Available at: https://www.tensorflow.org/api_docs/python/tf/keras/layers/LSTM

[Accessed September 2026].